

EXPERIMENT-4

Exploration vs. Exploitation Analysis using Multi-Armed Bandit Algorithms

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Course Name: REINFORCEMENT LEARNING

Slot: B

Aim

To analyze and compare **exploration and exploitation strategies** using a **Multi-Armed Bandit** algorithm.

Procedure

1. Define multiple bandit arms with different reward probabilities.
2. Implement an **ϵ -greedy strategy** for exploration and exploitation.
3. Run the bandit for a fixed number of rounds.
4. Record the rewards and arm selections.
5. Compare the **total and cumulative rewards** of exploration and exploitation.
6. Visualize the comparison using graphs.

Python code:

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

# =====

# EXPLORATION VS EXPLOITATION - MULTI-ARMED BANDIT

# =====

np.random.seed(42)
```

```
# Three bandit arms with different reward probabilities
```

```
arms = [0.3, 0.6, 0.8]
```

```
n_rounds = 100
```

```
# -----
```

```
# 1. Exploration: Epsilon-Greedy
```

```
# -----
```

```
epsilon = 0.2
```

```
counts_exp = np.zeros(3)
```

```
rewards_exp = np.zeros(3)
```

```
total_exp = 0
```

```
history_exp = []
```

```
for t in range(n_rounds):
```

```
    # Exploration
```

```
    if np.random.rand() < epsilon:
```

```
        arm = np.random.randint(3)
```

```
    # Exploitation
```

```
    else:
```

```
        arm = np.argmax(  
            rewards_exp / np.maximum(counts_exp, 1)  
        )
```

```
    reward = np.random.rand() < arms[arm]
```

```
    counts_exp[arm] += 1
```

```
    rewards_exp[arm] += reward
```

```
    total_exp += reward
```

```

    history_exp.append(total_exp)

# -----

# 2. Pure Exploitation

# -----

counts_exploit = np.zeros(3)
rewards_exploit = np.zeros(3)
total_exploit = 0
history_exploit = []

for t in range(n_rounds):

    # Select arm with highest estimated reward

    arm = np.argmax(

        rewards_exploit / np.maximum(counts_exploit, 1)

    )

    reward = np.random.rand() < arms[arm]

    counts_exploit[arm] += 1

    rewards_exploit[arm] += reward

    total_exploit += reward

    history_exploit.append(total_exploit)

# -----

# 3. Summary Table

# -----

summary = pd.DataFrame({

    "Method": [

        "Exploration ( $\epsilon = 0.2$ )",

```

```

        "Exploitation ( $\epsilon = 0$ )"
    ],
    "Total Reward": [
        int(total_exp),
        int(total_exploit)
    ],
    "Average Reward": [
        round(total_exp / n_rounds, 3),
        round(total_exploit / n_rounds, 3)
    ]
})

print("\n===== SUMMARY =====\n")

display(summary)

# -----

# 4. Arm Selection Comparison

# -----

arm_summary = pd.DataFrame({
    "Arm": ["Arm 1", "Arm 2", "Arm 3"],
    "True Reward Probability": arms,
    "Exploration Selections": counts_exp.astype(int),
    "Exploitation Selections": counts_exploit.astype(int)
})

print("\n===== ARM SELECTION COMPARISON =====\n")

display(arm_summary)

```

```

# -----

# 5. Total Reward Comparison

# -----

plt.figure(figsize=(8, 5))

plt.bar(

    ["Exploration", "Exploitation"],

    [total_exp, total_exploit]

)

plt.title("Exploration vs Exploitation")

plt.xlabel("Strategy")

plt.ylabel("Total Reward")

plt.grid(axis="y", linestyle="--", alpha=0.4)

plt.tight_layout()

plt.show()

# -----

# 6. Cumulative Reward Comparison

# -----

plt.figure(figsize=(9, 5))

plt.plot(

    range(1, n_rounds + 1),

    history_exp,

    label="Exploration ( $\varepsilon = 0.2$ )",

    linewidth=2

)

```

```
plt.plot(
    range(1, n_rounds + 1),
    history_exploit,
    label="Exploitation ( $\epsilon = 0$ )",
    linewidth=2
)

plt.title("Cumulative Reward Comparison")

plt.xlabel("Round")

plt.ylabel("Cumulative Reward")

plt.legend()

plt.grid(True, linestyle="--", alpha=0.4)

plt.tight_layout()

plt.show()
```

Output:

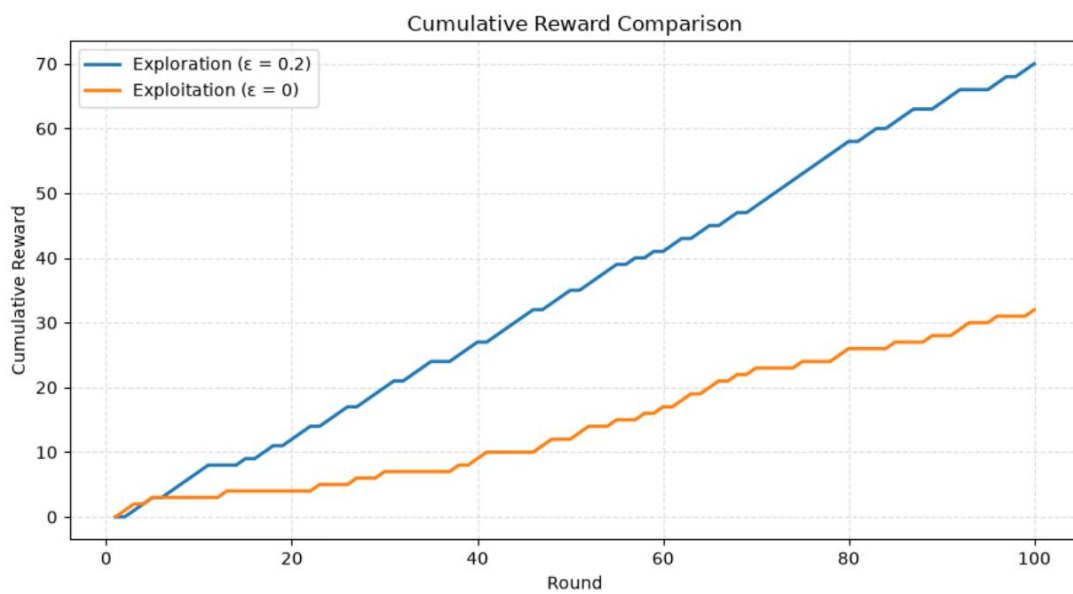
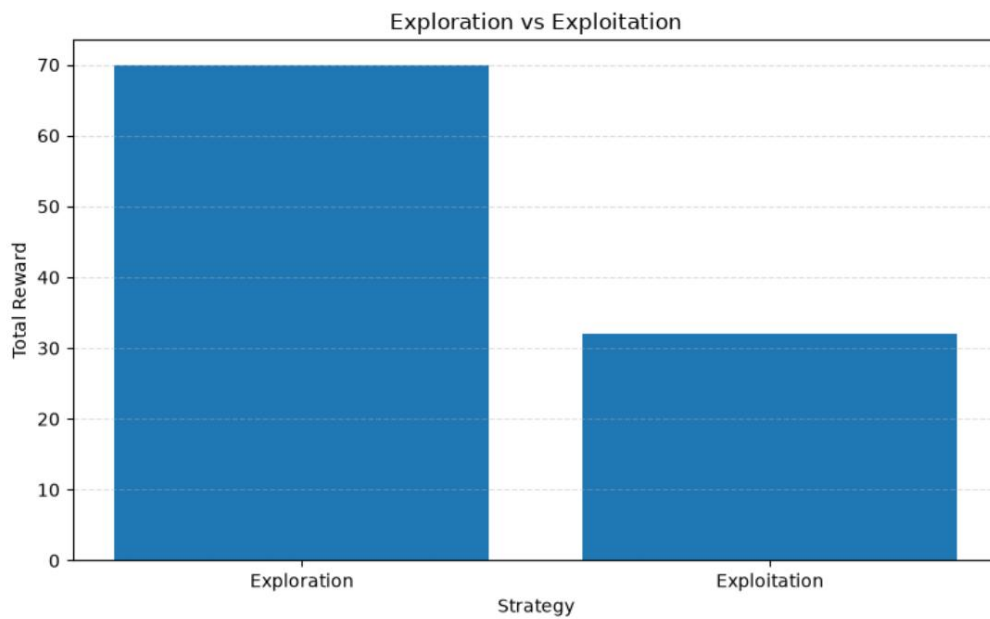
===== SUMMARY =====

	Method	Total Reward	Average Reward
0	Exploration ($\epsilon = 0.2$)	70	0.70
1	Exploitation ($\epsilon = 0$)	32	0.32

===== ARM SELECTION COMPARISON =====

	Arm	True Reward Probability	Exploration Selections	Exploitation Selections
0	Arm 1	0.3	8	100
1	Arm 2	0.6	9	0
2	Arm 3	0.8	83	0

Visualization:



Result

The **exploration and exploitation strategies were successfully implemented and compared**. The total reward, average reward, arm selections, and cumulative rewards were analyzed using summary tables and graphs.